The Cybersecurity Playbook

Foundations, Threats, and Defenses

The Scale of the Modern Cyber Threat

$10.5 Trillion

The projected annual global cost of cybercrime damages by 2025.

39 Seconds

The frequency of hacker attacks against connected systems.

95%

The percentage of security breaches caused by human error.

We are all targets. Modern threats combine physical and cyber methods to bypass layered defenses.

The Core Objective: Protecting the CIA Triad

Confidentiality

Protecting information from unauthorized access.

Breach Impact: Sensitive data exposed (e.g., SSN, financial info).

Availability

Guaranteeing authorized users can access systems when needed.

Breach Impact: Service outages and productivity loss.

Integrity

Ensuring information is accurate and protected from modification.

Breach Impact: Data altered or corrupted without authorization.

The Attacker’s Approach: Passive vs. Active

Passive Attacks

Silently monitors or intercepts data (e.g., Eavesdropping, Network Sniffing). The attacker is invisible; data flows normally but is copied.

Key Defense: Encryption

Active Attacks

Modifies, disrupts, or destroys data (e.g., Malware, DoS). The attacker interacts directly and causes visible damage.

Key Defense: Firewalls and IDS

The Golden Rule:
Active attacks change the system.
Passive attacks only watch it.

Categorizing the Malware Threat

	How it spreads	Requires User Action?	Main Goal
Virus	Infects host files	Yes (opening file)	Corrupt or replicate
Worm	Network auto-spread	No	Rapid propagation and resource consumption
Trojan	Disguises as legitimate software	Yes (installation)	Open a backdoor for other malware
Spyware	Silent installation	Sometimes	Steal passwords and financial data
Adware	Bundled downloads	Yes	Flood users with unwanted pop-up ads

The Ransomware Epidemic

$20 Billion

Cost to businesses globally in 2021.

Notorious examples: CryptoLocker, Petya, Locky.

1. Infect

Malware delivered via email or download.

2. Encrypt

Files and system resources are locked.

3. Demand Ransom

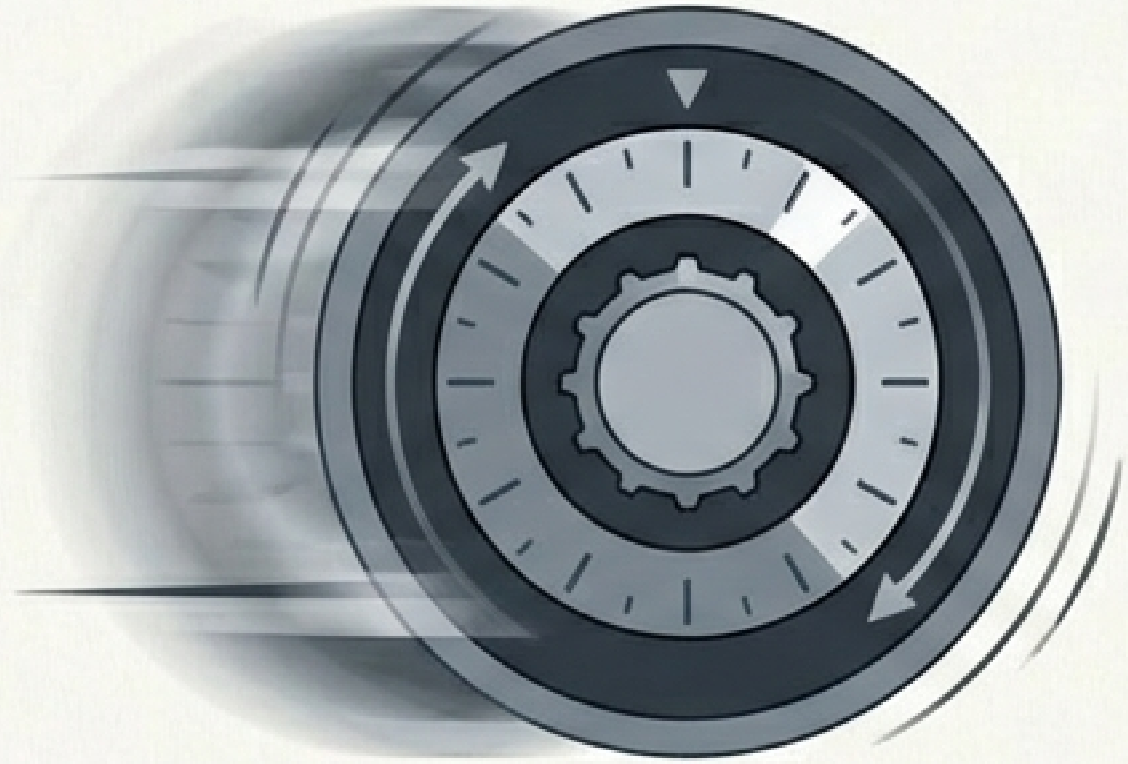
Attacker demands cryptocurrency for the decryption key.

4. Pay / Lose

Victim must pay the extortion fee or lose data permanently.

Breaking the Front Door: Network and Web Attacks

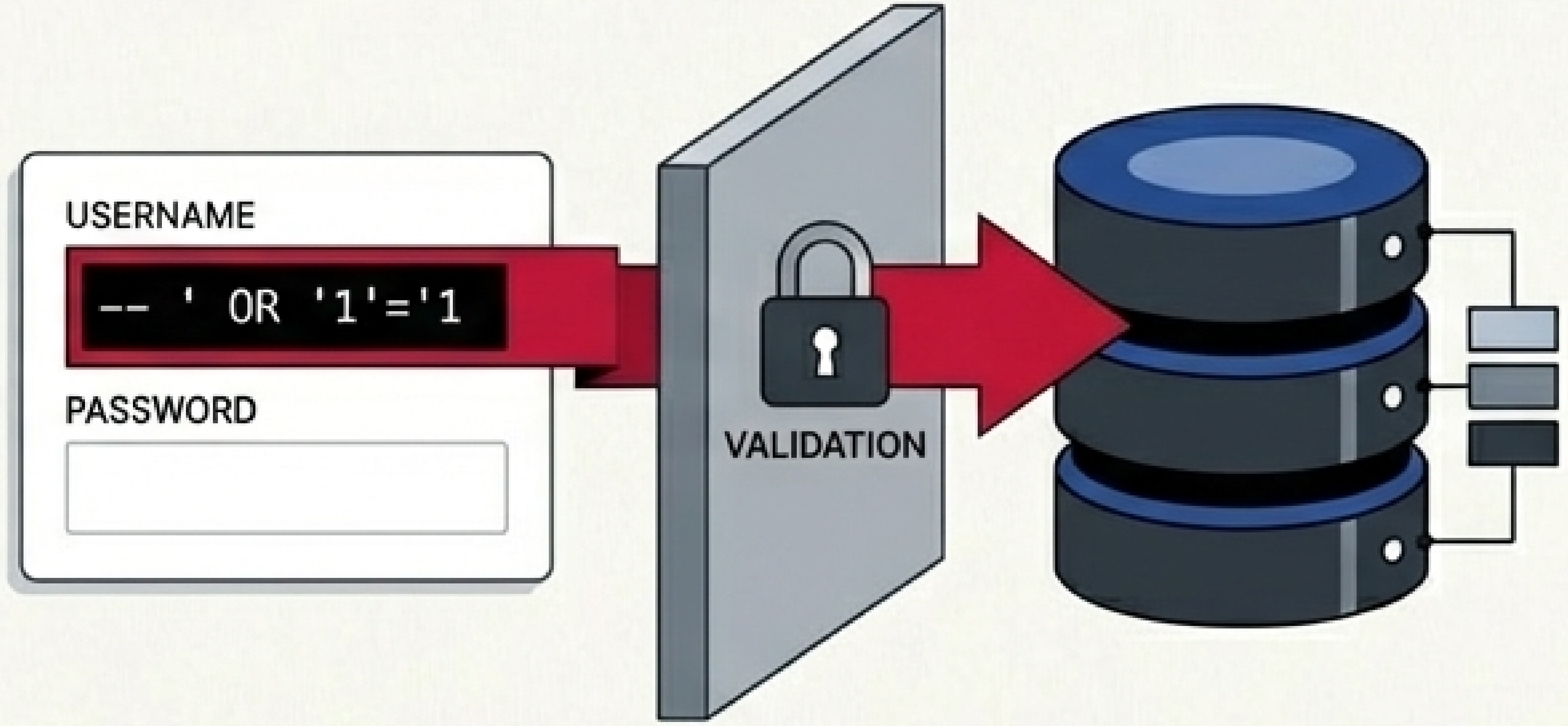
Brute Force



Automated trial-and-error tools attempt billions of guesses per second to crack passwords or encryption keys.

Targets: Login portals, SSH servers

SQL Injection



An attacker inserts malicious SQL code into an input field to manipulate the backend.

Forces a TRUE condition to bypass authentication.

The Invisible Interceptors

Man-in-the-Middle (MITM)

The attacker intercepts unencrypted data on public Wi-Fi, reads/modifies the traffic, and serves a fake reply.

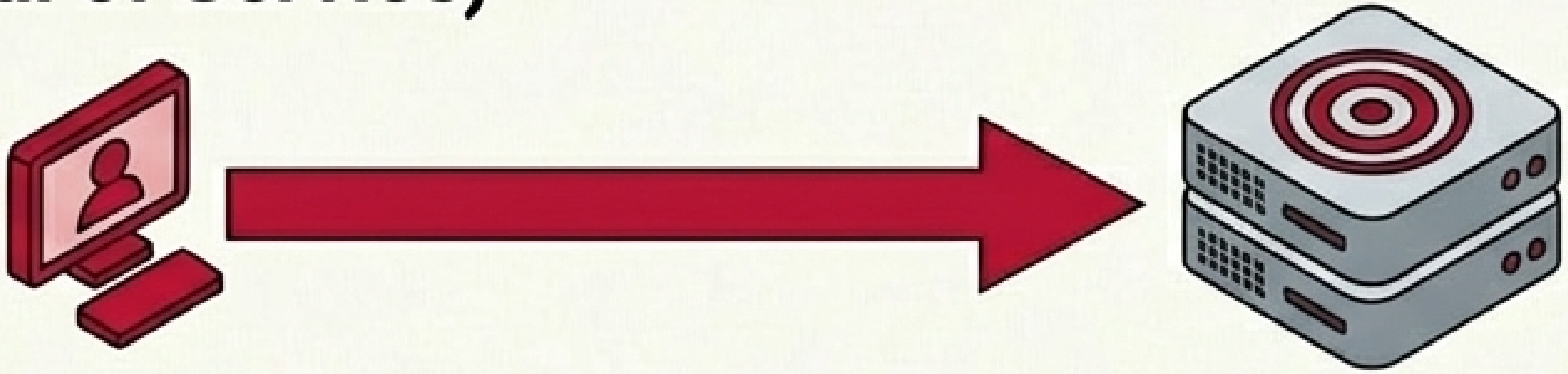
Cryptojacking: Warning Signs

- Extreme Heat
- Severe CPU Spikes
- Rapid Battery Drain
- Sluggish Performance

Criminals secretly use a device's CPU to mine cryptocurrency via hidden browser JavaScript or malware.

Disruption at Scale: DoS vs. DDoS

DoS (Denial of Service)

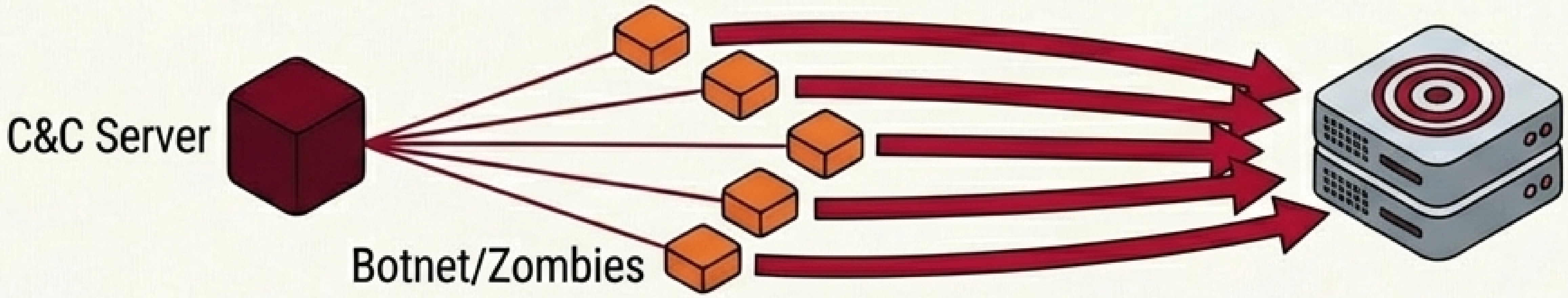


Single attacker machine flooding a target server with excessive traffic, blocking legitimate users.

Case Study: Mirai Botnet, 2016

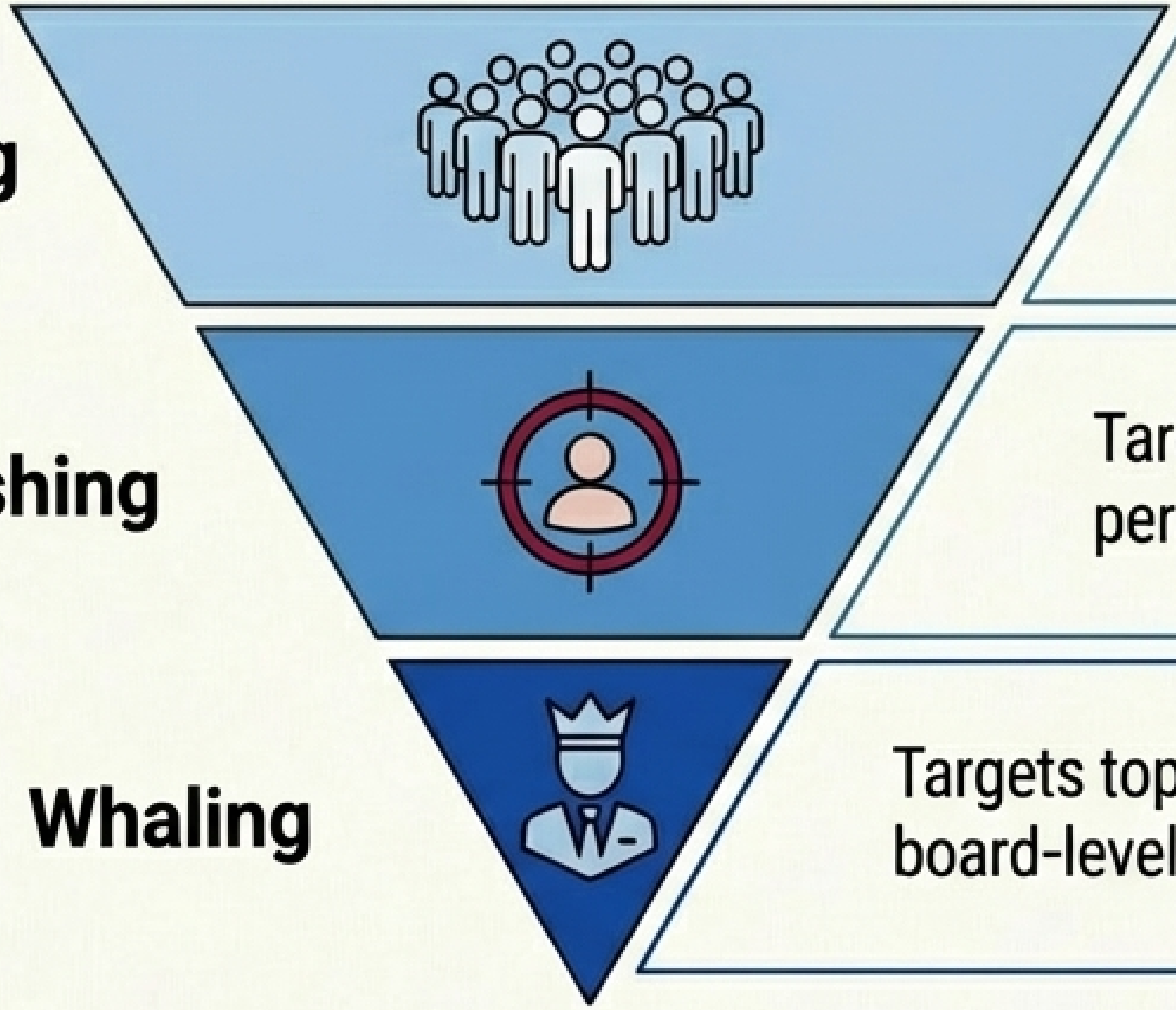
Enslaved 600,000 IoT devices (cameras, DVRs) to launch a 1 Terabit/second attack, taking down GitHub, Netflix, and Twitter.

DDoS (Distributed Denial of Service)



A Command-and-Control server commanding a massive botnet to attack simultaneously from multiple locations.

The Human Exploit: Phishing and Social Engineering



Generic Phishing

Mass-sent to thousands. Low effort, low success rate, but high volume yields results.

Spear Phishing

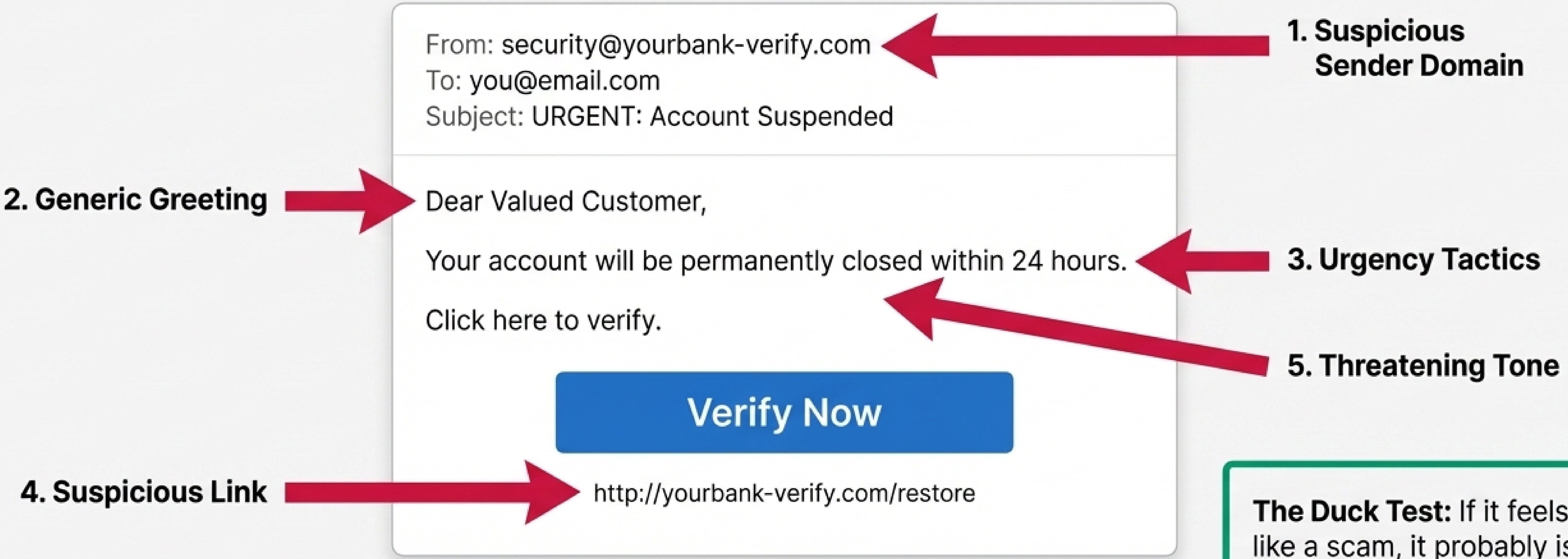
Targeted at a specific person using researched personal details. High success rate.

Whaling

Targets top executives (CEO, CFO) with urgent legal or board-level notices. Highest potential impact.

Social engineering manipulates people rather than technology, exploiting those who follow rules and trust authority.

Applying the Duck Test: Anatomy of a Phishing Defense



The Duck Test: If it feels like a scam, it probably is. Stop and verify through trusted channels.

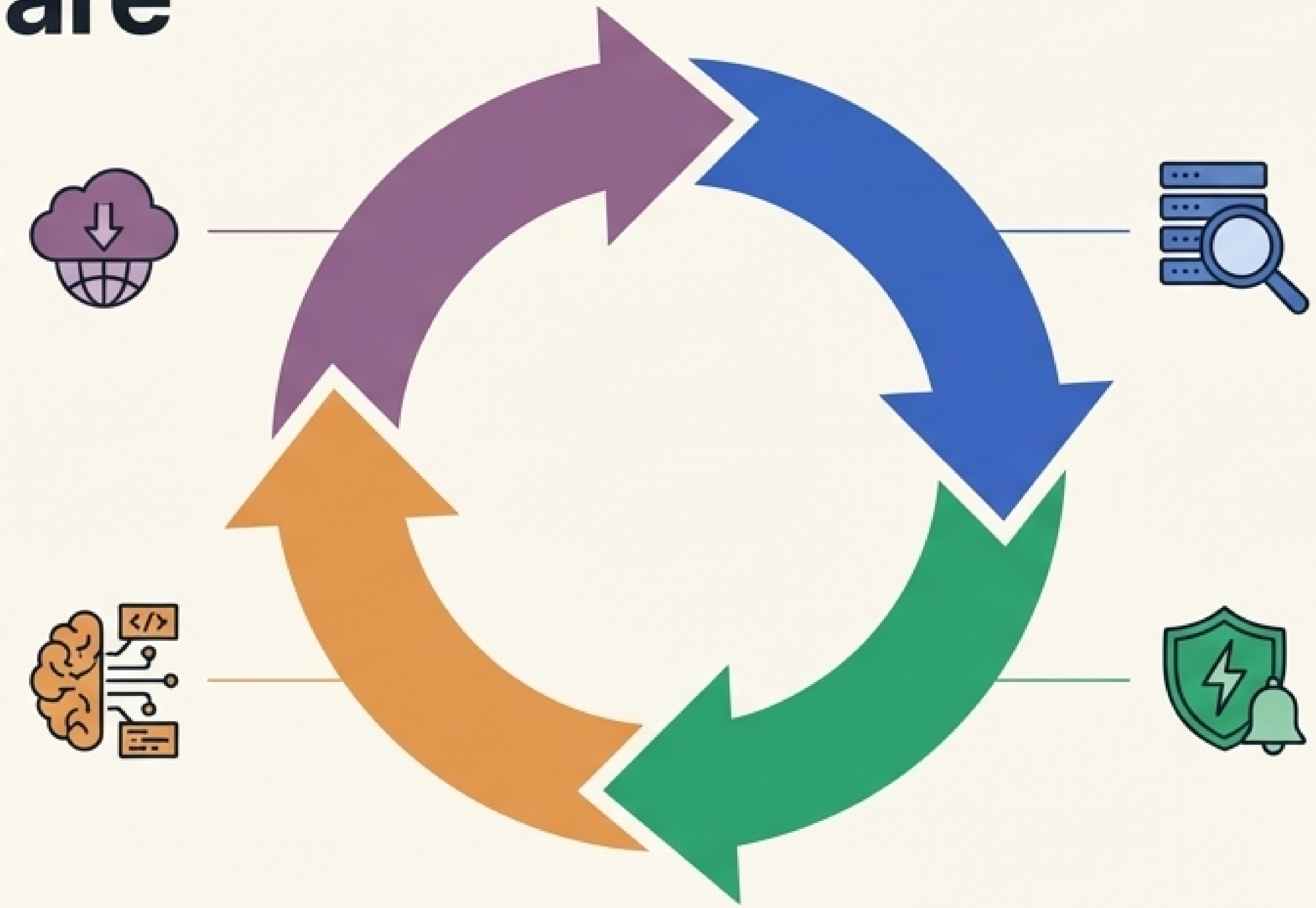
The Gatekeeper: Network Firewalls

Core Function: Monitors and controls incoming/outgoing traffic based on predetermined rules.

Note: Firewalls control what traffic enters; they do not catch everything. Defense-in-depth is required.

The Default DROP Rule:
If traffic is not explicitly permitted, it is denied.

Endpoint Protection: Antivirus and Anti-Spyware



1. Signature Scans

Checking files against known virus databases.

2. Real-Time Alerts

Catching threats in the background as they happen.

3. Behavioral AI

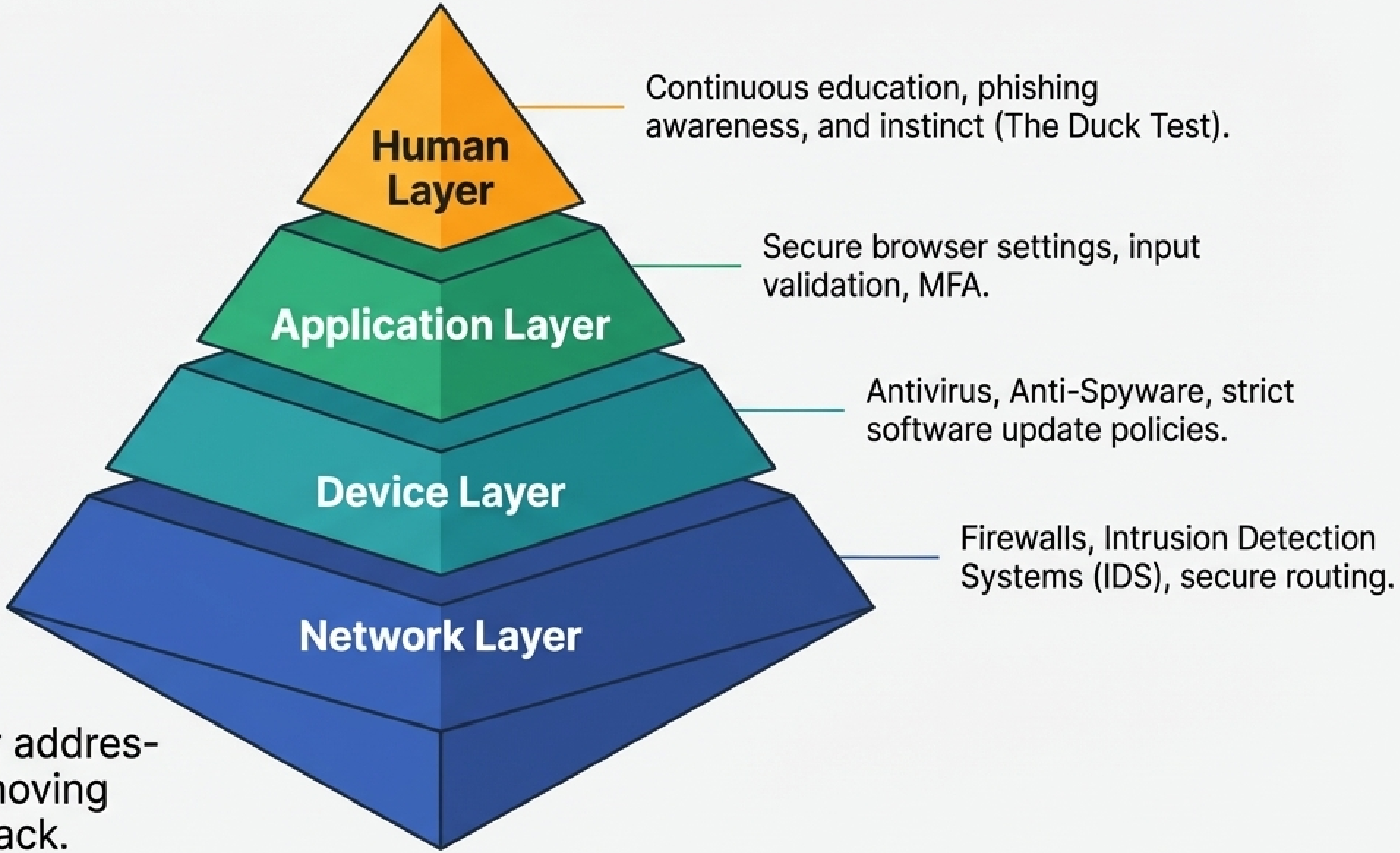
Identifying suspicious software actions at runtime.

4. Cloud Intel

Neutralizing emerging threats using global databases.

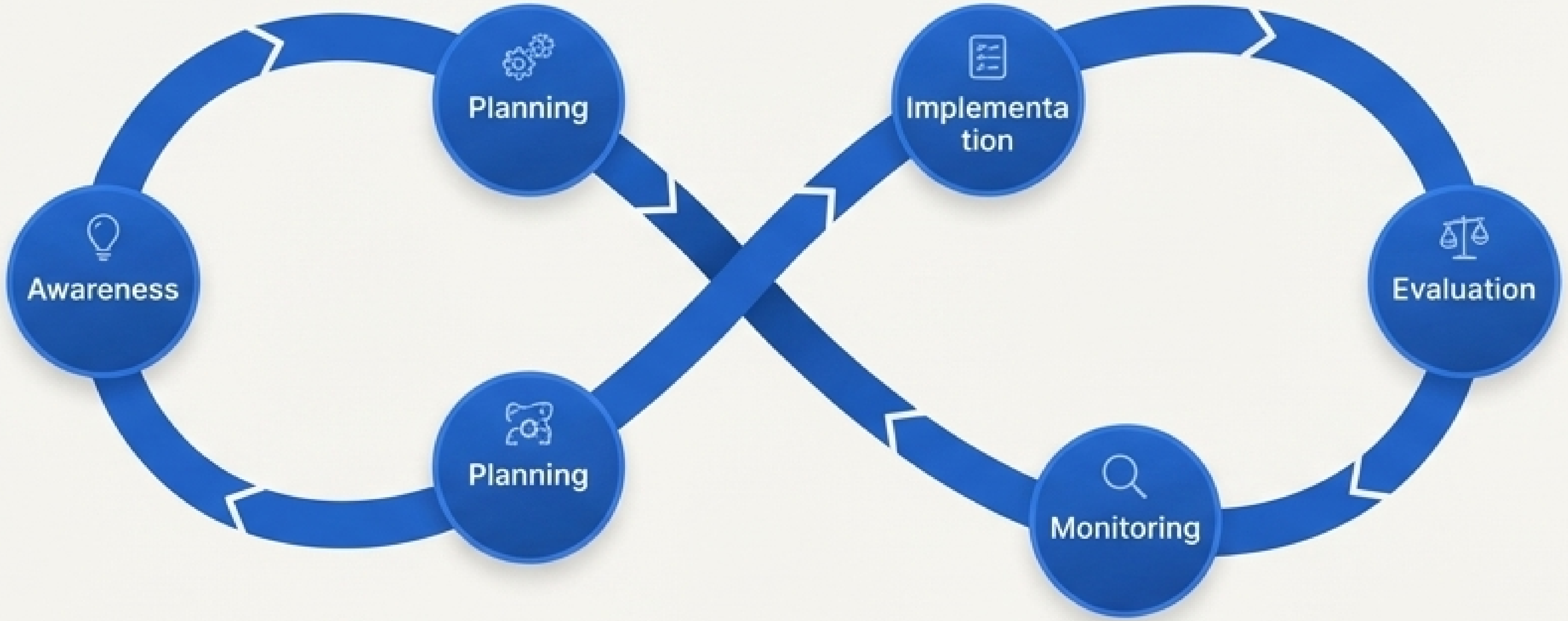
Firewall controls network traffic. Antivirus controls running files. Both are required.

The Layered Security Stack



Key Takeaway: Each layer addresses a different threat. Removing one weakens the entire stack.

The Continuous Cybersecurity Lifecycle



"The first step in securing a system is understanding how it can be attacked."